

Catchment zoning for freshwater conservation: refining plans to enhance action on the ground

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Summary

1. Recent advances in freshwater conservation planning allow addressing some of the specific needs of these systems. These include spatial connectivity or propagation of threats along stream networks, essential to ensure the maintenance of ecosystem processes and the biodiversity they sustain. However, these peculiarities make conservation recommendations difficult to implement as they often require considering large areas that cannot be managed under conventional conservation schemes (e.g. strict protection).

2. To facilitate the implementation of conservation in freshwater systems, a multizoning approach with different management zones subject to different management regimes was proposed. So far, this approach has only been used in *post hoc* exercises where zones were allocated using expert criteria. This might undermine the cost-effectiveness of conservation recommendations, because both the allocation and extent of these zones have never been optimized using the principles of systematic planning.

3. Here, we demonstrate how to create a catchment multizone plan by using a commonly applied tool in marine and terrestrial realms. We first test the capability of Marxan with Zones to address problems in rivers by using a simulated example and then apply the findings to a real case in the Daly River catchment, northern Australia. We also demonstrate how to address common conservation planning issues, such as accounting for threats or species-specific connectivity needs in this multizone framework, and evaluate their effects on the spatial distribution and extent of different zones.

4. We found that by prioritizing the allocation of zones subject to different management regimes, we could minimize the total area in need of strict conservation by a twofold factor. This reduction can be further reduced (threefold) when considering species' connectivity needs. The integration of threats helped reduce the average threats of areas selected by a twofold factor.

5. *Synthesis and applications.* Catchment zoning can help refine conservation recommendations and enhance cost-effectiveness by prescribing different management regimes informed by ecological needs or distribution of threats. Reliable information on these factors is a key to ensure soundness of planning. Freely available software can be used to implement the approach we demonstrate here.

Key-words: catchment management zone, connectivity, conservation planning, cost-effectiveness, critical management zones, freshwater focal areas, Marxan with Zones, systematic planning

Introduction

Freshwater conservation planning is a rapidly growing discipline (Collier 2011; Linke, Turak & Nel 2011). This

quick expansion is a response to the poor conservation status of freshwater ecosystems and biodiversity worldwide (Vörösmarty *et al.* 2010), the deficient consideration of freshwater biodiversity and conservation needs in existing protected areas (Nel *et al.* 2009) and the lack of methods to adequately address conservation planning for these

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systems (Linke, Turak & Nel 2011). Conservation of freshwater ecosystems has traditionally remained peripheral to conservation goals developed for terrestrial ecosystems, unless considered important for terrestrial biodiversity (Nel *et al.* 2007), partially due to the difficulty in addressing the peculiarities of these systems. Planning for conservation in freshwater ecosystems poses some unique challenges mainly derived from the lineal structure of river networks and the importance of connectivity for maintaining key ecological processes such as movement and migrations of biota or natural fluxes of energy and matter (Fausch *et al.* 2002). Moreover, effective conservation planning in freshwater ecosystems must consider the propagation of threats along stream networks (Linke, Turak & Nel 2011; Linke *et al.* 2012). The last few years have witnessed an outbreak of novel ways to integrate these special needs of freshwater systems into well-established systematic conservation planning methods previously developed and widely applied in marine and terrestrial environments (see Moilanen, Leathwick & Elith 2008; Hermoso *et al.* 2011; Hermoso, Kennard & Linke 2012; Nel *et al.* 2011 for some examples).

However, as a result of the unique spatial needs of freshwater ecosystems, conservation recommendations delivered by these novel applications of systematic planning approaches usually extend over large areas (Linke *et al.* 2007; Thieme *et al.* 2007; Moilanen, Leathwick & Elith 2008), which makes their implementation difficult. For example, a common solution to address the propagation of upstream threats into freshwater protected areas is to include whole (Linke *et al.* 2007; Thieme *et al.* 2007) or large portions (Hermoso *et al.* 2011; Hermoso, Ward & Kennard 2013) of the upstream catchment under the label of priority area for conservation (Fig. 1a). However, little is recommended in terms of the actual management regime required in those areas. This constrains the value of planning outcomes because these large areas cannot realistically be managed effectively under conventional conservation regimes (e.g. strict protection by designation as a National Park) due to potential conflicts with existing land uses. In an attempt to make conservation in freshwater ecosystems more practical, Abell, Allan & Lehner (2007) proposed a multizoning approach to help fulfil the spatial needs and ensure effective protection in a more flexible way (Fig. 1b). This zoning is composed of (i) 'freshwater focal areas', which are key areas for the protection of freshwater biodiversity, similar to protected areas in terrestrial or marine realms; (ii) 'critical management zones', as areas that need to be managed to maintain the ecological functionality of a focal area (e.g. connectivity to allow movement of individuals and gene exchange) and where uses that do not interfere with the purpose of this area are allowed; and (iii) 'catchment management zones' that link the entire upstream catchment to a critical management zone, where human uses are not constrained, but best practices (e.g. treatment of wastewater disposal, maintenance of riparian buffers in

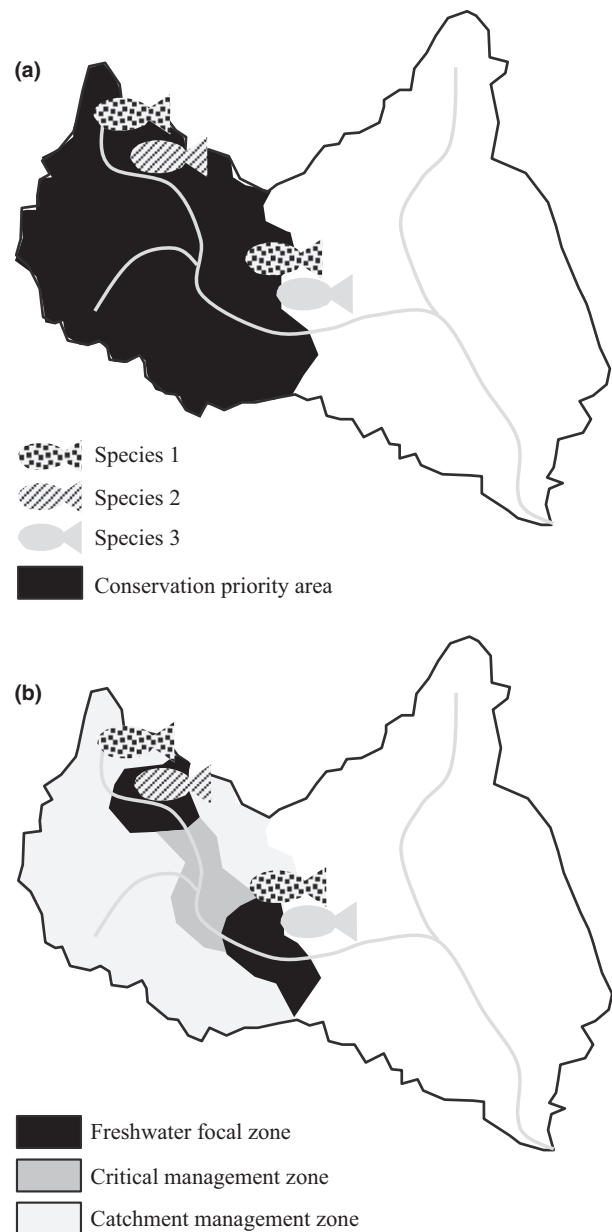


Fig. 1. Schematic representation of (a) conventional conservation priority area and (b) catchment zoning proposed by Abell, Allan & Lehner (2007) to protect three fish species. The conventional conservation priority area in (a) aims to encompass at least part of the distribution of all three species, including upstream contributing catchments, whereas the critical management zone in (b) aims to also facilitate connection between populations of species one.

good condition or by restricting the use of pesticides) are required (Fig. 1b). This approach is increasingly being accepted as an appropriate freshwater conservation framework (e.g. Linke, Turak & Nel 2011; Nel *et al.* 2011; Esselman *et al.* 2013), but has rarely been applied or tested and has not yet been integrated into systematic planning. Instead, past attempts at implementing the Abell, Allan & Lehner (2007) framework have all been conducted in a *post hoc* fashion (e.g. Thieme *et al.* 2007; Nel *et al.* 2011; Hermoso, Ward & Kennard 2013). Because they do not

explicitly incorporate complementarity and cost-effectiveness into the prioritization process (Margules & Pressey 2000), these *post hoc* approaches undermine the efficiency that made systematic frameworks popular.

Here, we demonstrate how to operationalize the Abell, Allan & Lehner (2007) zoning framework to create a systematic multizone conservation plan for river catchments. We use Marxan with Zones (Watts *et al.* 2009) for the first time in a freshwater context to create a catchment plan where different management zones are prioritized simultaneously. We integrate the spatial framework developed by Hermoso *et al.* (2011) to address connectivity in freshwater conservation planning into Marxan with Zones. We first demonstrate the capabilities of Marxan with Zones under this new spatial framework on a simulated example. We then test the approach using a case study in northern Australia (Daly River catchment) where we apply the Abell, Allan & Lehner (2007) framework to identify freshwater focal areas, critical management zones and catchment management zones for freshwater fish. We compare the results with those obtained using a traditional Marxan analysis, and we explore the effects of incorporating threat intensity and species-specific

connectivity needs on the spatial distribution and extent of different zones. These evaluations aim to further demonstrate how to address common conservation planning challenges. We conclude by providing recommendations to guide future applications of our approach that will help improve the design and implementation of cost-effective conservation plans for freshwater ecosystems.

Materials and methods

DEMONSTRATING THE USE OF MARXAN WITH ZONES IN RIVERS

To test the potential use of Marxan with Zones in a freshwater setting, we first simulated a simple case with a linear structure composed of ten consecutive planning units flowing from a headwater planning unit to a simulated outlet (Fig. 2a). We used this structure to build a connectivity file as usually done for Marxan applications in freshwater environments (Hermoso *et al.* 2011). This file differs from terrestrial and marine boundary files as it is made of all longitudinal connections between planning units. Penalties in the boundary file (Table 1) are distance weighted according to the distance between planning units along the river network [$penalty = distance\ (km)^{-1/2}$; Fig. 2a].

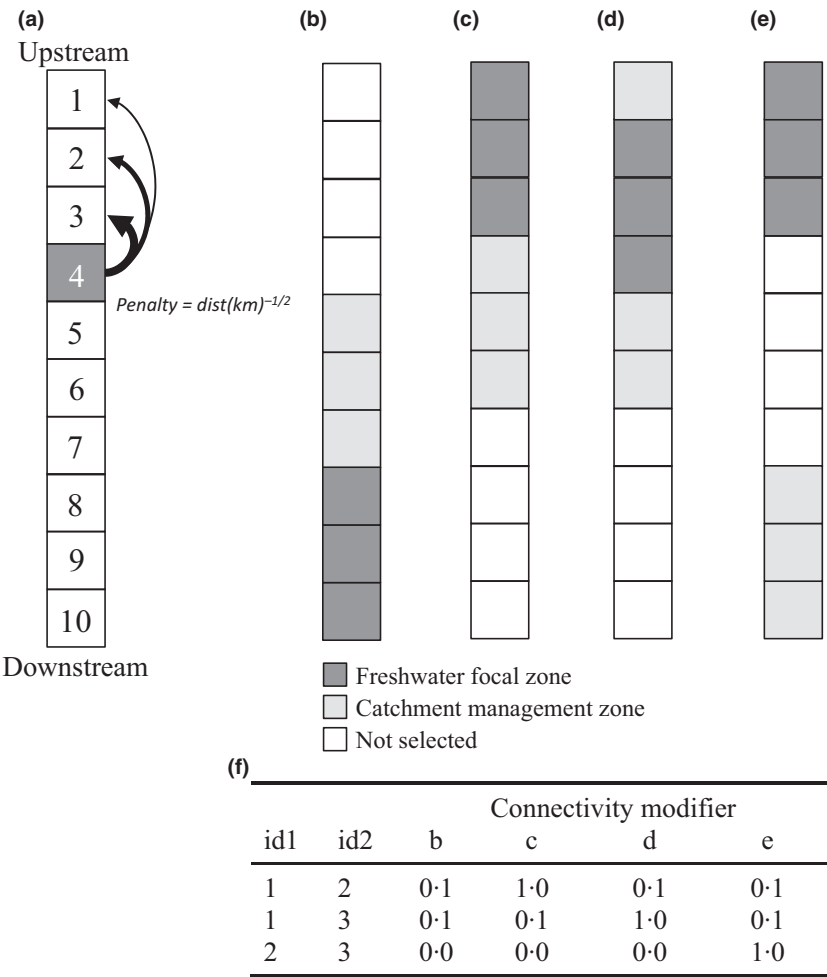


Fig. 2. Results from the simulation exercise using Marxan with Zones to identify conservation and buffer zones in a freshwater framework. (a) represents the spatial framework of the simulation exercise, with ten subcatchments running from one to ten and the way connectivity penalties were calculated (see Hermoso *et al.* 2011 for more detail). (b–e) show the spatial arrangement of freshwater focal zone (dark grey) and catchment management zone (light grey) obtained for the different configuration of the zone boundary file in (f).

Table 1. List of input files for Marxan with Zones and general usage

File name	Content	Description
Species	id, target*, spf*, name	List of conservation features, targets for each and weight to penalty for not achieving them (spf)
Planning units	id, cost*, status*	List of planning units, their cost and status
Species vs. planning units	species_id, planning_unit_id, amount	Definition of spatial distribution of conservation features
Boundary	id1, id2, boundary	Definition of connections between planning units and the penalty for missing them in a solution (boundary)
Zones	zone_id, zonename	Definition of number of zones and their names
Cost	cost_id, costname	Definition of different types of costs applicable to zones
Zone cost	zone_id, cost_id, modifier*	Weight of cost in each zone by the modifier. If a cost does not apply in a zone, the modifier should be set to zero
Zone boundary	zone_id1, zone_id2, connectivity modifier [†] [see Table 3]	Definition of spatial arrangement between zones (scenarios one–three)
Zone target	zone_id, feature_id, target [†] [see Table 2], type	Definition of amount of each feature to be achieved within each zone (scenario three)

*Indicates parameters that can be adjusted in Marxan with Zones, but that were kept constant in this study for the sake of demonstration (target = 200 km²; spf = 10; cost = 1; status = 0; zone cost modifier = 1).

[†]Indicates parameters that were modified across scenarios (see Table 3 for the different configurations of the zone boundary file and Table 2 for the different configurations of the zone target file). The remaining parameters cannot be modified in Marxan (mostly names and ids) and were kept constant across scenarios.

In this example, we assumed planning units to have a regular shape with a total length of river within each of 10 km, so, for example, the penalty for including planning unit four but not three in the solution would be 0.32 (penalty = $10^{-1/2}$). This penalty decays exponentially with distance between planning units, so the farther two planning units are apart, the lower the penalty that would apply if not selected together. To keep this test as simple as possible, we created two zones: a conservation zone and a buffer zone, respectively. We simulated the distribution of three conservation features, which occurred in all planning units. We also assumed equal cost for all planning units. For the sake of demonstration and simplicity, we used two zones in this example: freshwater focal zone and catchment management zones. A target frequency of occurrence of 25% of each conservation feature's distribution was set for all conservation features and allowed Marxan with Zones to achieve 75% of the target for two conservation features within the freshwater focal zone and 25% in the catchment management zone, and 25% of the target of the third species within the freshwater focal zone and 75% within the catchment management zone. The use of these targets was set for demonstration purposes only and should be adjusted according to the goal of each management zone. For example, given the primary conservation focus of freshwater focal zones, most of the targets could be achieved within them to ensure effective protection. However, these can be modified to account for species-specific needs (see the case study below). The allocation of targets was set in the zone target file (Table 1), where the distribution of targets within each zone can be specified for each species (Watts, Steinback & Klein 2008). We finally tested four different configurations of the zone boundary file (Fig. 2b–f) following recommendations in Watts, Steinback & Klein (2008). This file is commonly used to specify how different zones should be arranged spatially (either connected or disconnected from other zones; Table 1). With these tests, we wanted to explore the effect of different configurations of the zone boundary file for lineal systems and to help guide the calibration of a more complex zone boundary file for application in our real-world case study in the Daly River catchment.

DALY RIVER: SPATIAL FRAMEWORK AND BIOLOGICAL DATA

We used the Daly River catchment in northern Australia (Fig. 3) as an example to demonstrate the application of Marxan with Zones. The Daly River encompasses 53 000 km² and is in relatively good environmental condition compared to other major rivers in Australia, but there is considerable pressure for further agricultural development and water demand (Chan *et al.* 2012). We derived 865 subcatchments from a 9-s digital elevation model (ANU Fenner School of Environment and Society and Geoscience Australia, 2008) in ARCGIS 10.1 (ESRI 2011) to use as planning units. Each planning unit included the portion of river length between two consecutive nodes or river connections (8.0 km on average) and its contributing area (66.1 km² on average), representing an appropriate grain size of planning units for freshwater conservation planning (Hermoso & Kennard 2012). We sourced the spatial distribution of 45 freshwater fish species (Hermoso *et al.* 2015) from Kennard (2010). This data base contained continuous predictions of spatial distribution for 104 freshwater fish species across northern Australia derived from multivariate adaptive regression splines models (Leathwick *et al.* 2005) at a fine scale (average area of predictive polygons was 3.6 km²). The predictive model was built on a data set of 1609 presence-only records and was validated using an independent data set of 719 presence-absence records (see Hermoso, Kennard & Linke 2012 for more details on predictive models).

MANAGEMENT ZONES AND CONSERVATION SCENARIOS

To prioritize a management plan as suggested by Abell, Allan & Lehner (2007), we created three zones, equivalent to freshwater focal areas, critical management zones and catchment management zones. We used the framework to address connectivity in freshwater conservation planning proposed by Hermoso *et al.* (2011) as described above using the real stream and subcatchment

topology in the Daly River in this case. We used the boundary zone and zone target files (Watts, Steinback & Klein 2008) to guide Marxan with Zones, how the different zones should be arranged spatially and where conservation targets could be achieved according to the role each zone plays (Abell, Allan & Lehner 2007). We used three different conservation planning scenarios to explore the effect of different constraints such as threat intensity and species-specific connectivity needs (see below). Given our special interest in exploring the effect of boundary zone and zone target files, we set the remaining parameters constant across scenarios (Table 1). We set a constant conservation target of 200 km² for all species across all scenarios for demonstration purposes. Better ecologically informed targets would be needed when applying the method demonstrated here to develop real-world conservation recommendations. Our approach to conservation target setting is conservative because it represents the total distribution ranges for the nine rarest species in the catchment and one-third of the total distribution across all species on average (Appendix S1 in Supporting Information). We also set a constant cost for all subcatchments (cost = 1 for all subcatchments) and applied a high species penalty factor (spf) to ensure all species achieve their representation target (spf = 1). The

Table 2. Proportion of the conservation target used to build the zone target file used for scenario three. We allowed representation targets to be achieved in different zones depending on each fish species' mobility (see Appendix S1 for more detail on each species' mobility). For example, the critical management zone could contribute with up to 40% of the target for high-mobility species but none for low-mobility ones

Species' mobility	Zone		
	Freshwater focal zone	Critical management zone	Catchment management zone
High	0.50	0.40	0.10
Intermediate	0.85	0.10	0.05
Low	1.00	0.00	0.00

Table 3. Configuration of the zone boundary file for the three alternative scenarios tested in this study. The combination of positive and negative weights was necessary to achieve the desired spatial arrangement of zones. Note that the values need to be calibrated for each case study. Negative values were found necessary to achieve connectivity either within or between zones after calibration following recommendations by Watts, Steinback & Klein (2008)

zone_id1	zone_id2	Scenario 1	Scenario 2	Scenario 3
1	1	-0.2	-0.1	-0.1
1	2	0.0	0.2	0.2
1	3	-0.5	-0.4	-0.4
1	4	0.5	0.4	0.4
2	2	-0.8	-0.6	-0.5
2	3	-0.9	-0.3	-0.5
2	4	-0.8	-0.7	-0.6
3	3	-0.6	-0.3	-0.5
3	4	-1.0	-0.7	-0.5
4	4	-0.8	-0.3	-0.5

importance of achieving targets is weighted by the spf, so the higher the spf, the less chance of missing some species from the plan.

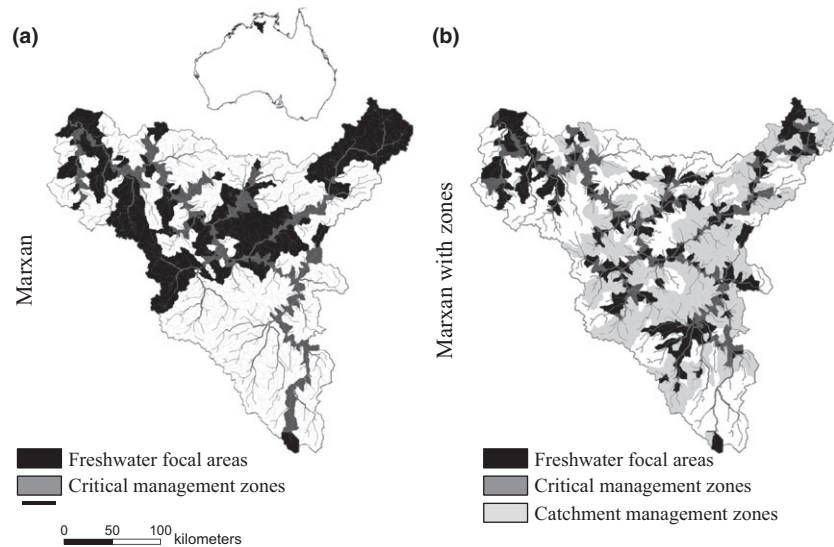
Scenario 1: Catchment zoning

In our first conservation scenario, we calibrated the weights in the zone boundary file to arrange zones spatially in a similar way as proposed by Abell, Allan & Lehner (2007). The spatial arrangement of zones that we sought was as follows: core conservation areas or freshwater focal areas connected through a critical management zone and buffered upstream by catchment management zones. We used the zone target file to ensure representing species mainly within freshwater focal areas (90% of representation targets) as these would be mostly devoted to conservation. Critical management zones and catchment management zones would contribute to the remaining 10% representation of targets (5% each) while enhancing connectivity and accounting for upstream threats. To explore the difference in the spatial allocation of priority areas with respect to traditional recommendations, we ran the same conservation planning prioritization (constant target = 200 km² for all species and cost = 1 for all subcatchments) in Marxan (Ball, Possingham & Watts 2009). Conservation costs are an important component of conservation planning because they can significantly influence the extent and allocation of priority areas for conservation (see Carwardine *et al.* 2008 for an example). In this scenario, we maintained cost constant for demonstration purposes and better estimates would be needed to warrant soundness and efficiency of conservation planning outcomes in real case studies. All subcatchments identified as priority areas in this analysis were labelled as freshwater focal zone. To compare the efficiency at identifying critical management zones identified in Marxan with Zones and Marxan, we then manually selected all catchments that connected to priority areas identified by Marxan as per existing studies. In these analyses, critical management zones are visually identified after freshwater focal zones have been prioritized to ensure full connection between freshwater focal zones (see Hermoso, Kennard & Linke 2012 for an example).

Scenario 2: Accounting for threats in catchment zoning

To account for feasibility of conservation and conservation costs derived from threat reduction, we included estimates of threat intensity across the catchment. Areas under high threat are less suitable for conservation because they would need additional conservation actions (e.g. eradication of invasive species or restoration of habitat quality) to ensure the threat to biodiversity is adequately addressed. We used the proportion of each subcatchment under grazing pressure, a widely recognized threat to freshwater ecosystems, as an estimate of their threat intensity for freshwater ecosystems (data sourced from the Department of Agriculture, Fisheries and Forestry, <http://www.daff.gov.au/abares/aclump/pages/about-aclump.aspx>; accessed July 2014). Given that real estimates of potential conservation management costs are currently unavailable, we used threat intensity as a surrogate for cost in the prioritization process similar to Linke *et al.* (2012). Marxan with Zones allows giving different costs to each zone to account for the different conservation requirements or socio-economic constraints imposed (Watts, Steinback & Klein 2008; Klein *et al.* 2009). We finally used the same zone target

Fig. 3. Spatial distribution of priority areas in the Daly River catchment obtained for scenario one with no consideration of threats and species-specific connectivity needs. (a) Best solution obtained from Marxan (dark grey) and *post hoc* addition of critical management zones (light grey) and (b) spatial arrangement of zones obtained from Marxan with Zones. Inset map shows the location of the Daly River catchment in northern Australia.



configuration as in scenario one (90% of targets within freshwater focal zones and 5% within critical management zone and catchment management zone).

Scenario 3: Accounting for species-specific connectivity needs and threats in catchment zoning

The long-term persistence of species within conservation priority areas will depend on the capacity to maintain key ecological processes that sustain them (Linke, Turak & Nel 2011). Maintaining conditions for unimpeded movement of species is important to ensure connectivity between different populations and completing their ecological needs (e.g. migrations between freshwater and downstream estuarine and coastal areas for diadromous species). We further wanted to integrate in our river zoning the different role that each zone has in maintaining ecological needs for species and securing their persistence. For example, the role of the critical management zone would be more important for highly mobile species, which might spend most of their life cycle within these critical management zones. Ensuring maintenance of connectivity along these zones would be critical to warrant conservation efficacy for these highly mobile species, for example. On the other hand, freshwater focal areas might play a more important role for species with low-mobility needs. Here, we accounted for the mobility of each species (Appendix S1) by modifying the zone target file to allow a larger proportion of the representation target to be achieved in the critical management zone and a lower proportion in focal (high-mobility focal 50%, 40% critical, 10% catchment compared to medium focal 85%, 10% critical, 5% catchment; Table 2). In this way, we wanted to account for the special contribution that the critical management zone would make to maintaining populations of highly mobile species. Conversely, we assumed that species with low mobility would only need to be represented within focal freshwater zones. We classified each species as high, intermediate and low mobility (Appendix S1) based on information in Pusey, Kennard & Arthington (2004) and adapted the zone target file accordingly (Table 2). For this scenario, we also included threats in the same way as in scenario two.

We ran the optimization algorithm 100 times (4 million iterations each) with a constant connectivity strength modifier (CSM = 1; Hermoso, Kennard & Linke 2012). The CSM controls

for the importance of connectivity (the higher the CSM, the more the connectivity) in the solutions, and it must be calibrated as shown by Hermoso *et al.* (2011) because excessively high CSM values might undermine the efficiency of solutions (e.g. larger areas than needed or affordable being selected to maximize connectivity). We retained the best solutions from Marxan with Zones for each scenario for comparative evaluation. We compared the extent (area measured in km²) and spatial allocation of each zone across scenarios to explore the effects of considering different constraints and ecological requirements. We also calculated the average threat intensity across subcatchments within each zone for further comparisons between planning scenarios.

Results

SIMULATION TEST

The different configurations of the zone boundary file led to differences in the spatial arrangement of zones in the simple simulation test (Fig. 2). The catchment management zone could be allocated either upstream or downstream of the freshwater focal zone by adjusting the connectivity modifier between freshwater focal areas and the 'available zone'. This available zone is used in Marxan with Zones to refer to areas that do not fall within any of the specified zones. Low connectivity modifier values produced a solution where the catchment management zone was located in upstream freshwater focal areas, as proposed by Abell, Allan & Lehner (2007). On the other hand, when all the other connectivity modifier scores were kept constant, high values of the connectivity modifier moved the allocation of the catchment management zone to downstream freshwater focal areas. By increasing the connectivity modifier between catchment management zone and available zone, we managed to completely buffer both upstream and downstream freshwater focal areas with the catchment management zone (Fig. 2b–d). Moreover, the zone boundary file could also be used to

segregate zones spatially, forcing complete disconnection between freshwater focal areas and catchment management zones by increasing the connectivity modifier between both management zones (Fig. 2e).

SCENARIO ONE: CATCHMENT ZONING

Priority areas identified with Marxan (Fig. 3a) and Marxan with Zones (Fig. 3b) were distributed across the whole catchment in similar tributaries and avoiding the main channel of the Daly River. Despite this similarity in the spatial distribution of priority areas, there was a significant reduction of 42% in the extent of freshwater focal areas identified using Marxan with Zones compared to Marxan (Fig. 4) and a significant change in the spatial allocation of these areas with only 22% of subcatchments in both solutions (Fig. 3a). While Marxan required selecting large areas to secure high spatial connectivity (Fig. 3a), Marxan with Zones was more efficient and identified only a small proportion of all selected subcatchments (28.7%) as freshwater focal areas, while the remaining subcatchments selected were labelled as either critical or catchment management zones.

This reduction in total spatial extent translated into an overall improvement in efficiency of 58% as both solutions achieved the targets for all the species. The three different zones in the solution from Marxan with Zones were arranged spatially to minimize the total area to be managed. For example, the extent of the critical management zone identified using Marxan with Zones would be 17% smaller than subcatchments needed to connect all freshwater focal areas identified with Marxan (the later

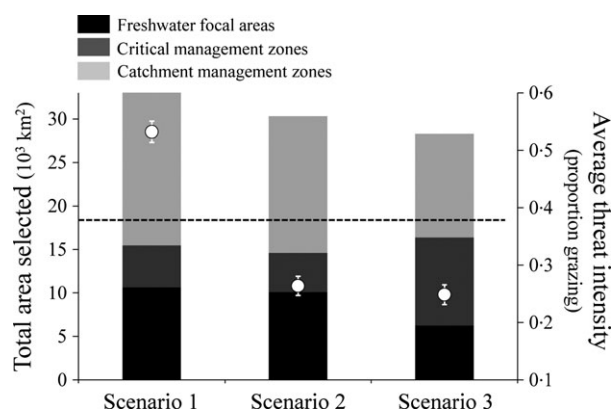


Fig. 4. Total area selected within each zone (bars) and average ($\pm$ SE) threat intensity of subcatchments included in the solution for each scenario (white circles). Threat intensity was estimated as the average proportion of subcatchment devoted to grazing, one of the main threats in the Daly River catchment used as the case study here. The dotted line depicts the total area selected by Marxan without consideration of threats or species' connectivity needs (planning assumptions equivalent to scenario one in Marxan with Zones, where no threats or species' mobility were addressed).

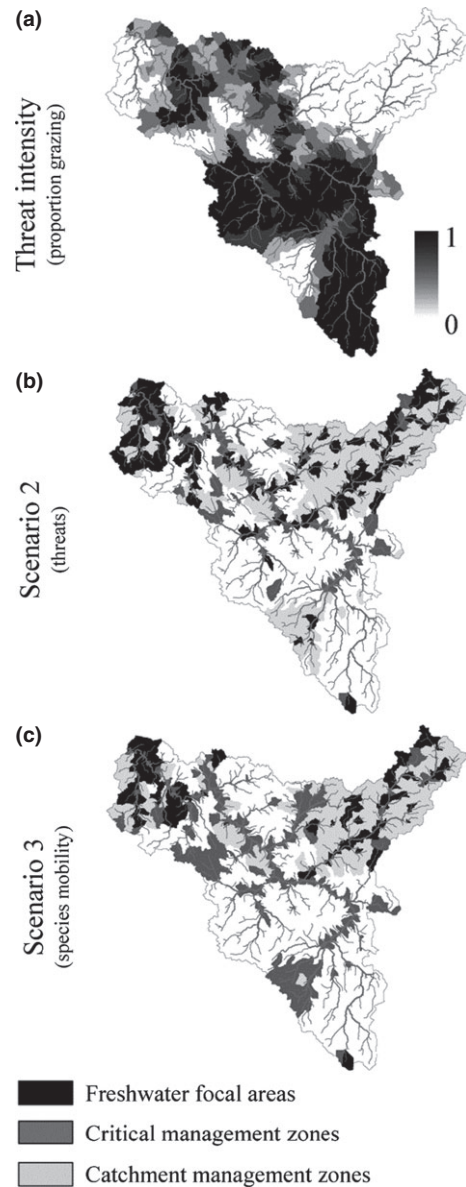


Fig. 5. (a) Spatial distribution of threat intensity (proportion of subcatchments devoted to grazing, pale-dark grey indicates low-high values, respectively) in the Daly River catchment, (b) spatial arrangement of zones for scenario two including threat intensity and (c) scenario three also considering species' mobility.

identified in a *post hoc* analysis to try to connect all subcatchments included in the solution).

SCENARIO TWO: ACCOUNTING FOR THREATS IN CATCHMENT ZONING

The spatial arrangement of zones changed significantly when we considered the distribution and intensity of threats in the catchment (Fig. 5a,b). Highly disturbed zones in the upper catchment (Fig. 5a) that were selected in scenario one were replaced now with other areas in the lower catchment where the threat intensity was lower, leading to a 50% reduction in average threat intensity

within selected subcatchments (Fig. 4). This change also brought a significant reduction in the extent of the catchment management zone (21.4% smaller than in scenario one; Fig. 4), while the other zones remained almost invariant with respect to scenario one. Importantly, the average threat intensity of the selected subcatchments was substantially lower than in scenario one (Fig. 4).

SCENARIO THREE: ACCOUNTING FOR SPECIES-SPECIFIC CONNECTIVITY NEEDS AND THREATS IN CATCHMENT ZONING

The consideration of species-specific connectivity needs in Marxan with Zones led to a further change in the relative extent of each zone (Fig. 4) and their spatial arrangement compared to scenario two (Fig. 5c). These changes were mainly related to the ability to achieve different proportion of species' targets across zones depending on each species' mobility. The reduction in the extent of freshwater focal areas compared to scenario two (about 38% smaller) translated into a twofold increase in the extent of critical management zones and 24% decrease in the extent of the catchment management zone (Fig. 4). In total, the consideration of ecological needs produced a 7% decrease in total area to be managed compared to scenario two and 20% reduction in area compared to scenario one. It also led to virtually the same average threat intensity within selected subcatchments compared to scenario two, and a reduction of 53% compared to scenario one. Critical management zones were not only associated with connecting freshwater focal areas, but also creating corridors to some headwater areas that could be important for highly mobile species. This was achieved by increasing the extent of critical management zones by 55%, but reducing the extent of focal freshwater areas and catchment management zones by 38% and 24%, respectively, compared to scenario two.

Discussion

This is the first study demonstrating a systematic multizone conservation management plan for freshwater biodiversity that combines the Abell, Allan & Lehner (2007) freshwater conservation framework with Marxan with Zones (Watts *et al.* 2009), a publicly available systematic planning tool. This is also the first time different management zones have been prioritized simultaneously in a riverine context to enhance cost-effectiveness of their spatial extent and arrangement. Our recommendations should help guide future applications of this software to address the commonly claimed need for flexible conservation recommendations in freshwater environments (Linke, Turak & Nel 2011; Nel *et al.* 2011; Esselman *et al.* 2013) and strengthen the value of conservation planning outcomes.

By integrating the spatial framework developed by Hermoso *et al.* (2011) for freshwater conservation planning in Marxan with Zones, we were able to arrange management zones spatially according to the special needs

of these systems (Abell, Allan & Lehner 2007). As in Marxan, the original applications of Marxan with Zones were initially applied to marine (Klein *et al.* 2009) and more recently terrestrial realms (Levin *et al.* 2013). These applications demonstrated how to arrange zones to minimize potential conflicts between different uses (e.g. extractive uses such as fisheries or logging) and enhance the capacity of reserves to maintain marine and terrestrial biodiversity (Klein *et al.* 2009). Here, we have used the capability of Marxan with Zones to create buffering zones in combination with the special freshwater spatial framework to arrange spatially the three different zones proposed by Abell, Allan & Lehner (2007). Ours are general management zones with no prescription of allowed or prohibited uses as we only included biodiversity targets in our analyses. Following Abell, Allan & Lehner's recommendations, intensive uses (such as river damming or water abstraction) should be avoided from freshwater focal areas, while certain uses could be allowed within critical management zones and especially catchment management zones if they are compatible with the main aim of each zone (Abell, Allan & Lehner 2007) (e.g. sustainable rotational grazing practices, Adams, Pressey & Stoeckl 2012). In a broader context, our work should also guide future zoning in freshwater ecosystems independently of the zone structure that would likely be implemented to address different planning needs not necessarily constrained to the one proposed by Abell, Allan & Lehner (2007). The number of zones, their aim and how they need to be spatially related to each other should be agreed with stakeholders to account for their specific needs. For example, we have only included zones with conservation purposes, but zones could be planned to account for different extractive or recreational uses as well as conservation as done in other realms (Klein *et al.* 2009). For example, we have shown how the zone boundary file could be used to not only spatially aggregate zones along the river network, but also disaggregate them if required. This could be used, for example, to plan the allocation of development zones for impacting uses (e.g. hydropower, water abstraction, fisheries) and minimize their potential impacts on conservation zones.

Previous conservation recommendations for freshwater ecosystems lacked an objective and systematic way of allocating different management zones and relied mainly on either visual inspections or *post hoc* analyses. Most of conservation planning recommendations either produced binary recommendations (protection or no protection) or identified different management zones using *post hoc* approaches. The first and most common approach is quite demanding in terms of the area required to represent not only biodiversity, but also processes such as migrations (Rivers-Moore, Goodman & Nel 2011), aquifer recharge (Nel *et al.* 2011) or resilience to climate change (Bush *et al.* 2014). A large extent of conservation recommendations could raise opposition among stakeholders and undermine the value of conservation recommendations.

For this reason, a more flexible approach with different zones is desirable (Abell, Allan & Lehner 2007; Linke, Turak & Nel 2011; Nel *et al.* 2011) but had seen little development to date. Previous implementations of this approach (e.g. Thieme *et al.* 2007; Nel *et al.* 2011; Hermoso, Ward & Kennard 2013) were done using *post hoc* analyses that undermine the efficiency of solutions obtained from systematic methods. Thus, multiple zoning should be prioritized simultaneously in order to identify the most efficient spatial configuration of all zones. As we demonstrate here, the allocation of any zone and extent are dependent on the constraints imposed to the other. For example, the allocation of freshwater focal areas will be constrained by our capacity to adequately manage upstream subcatchments included in the catchment management zone. They should also be allocated in places where the efforts of maintaining connectivity are minimized, which then minimize the extent of the critical management zone. An example of the influence of some zones on the allocation of the others is the results from our scenario two. The spatial allocation of freshwater focal areas changed towards less disturbed tributaries probably not only because of the potential *in situ* impact of threats, but also to minimize the management effort in the upstream catchment management zones. By shifting efforts towards tributaries in better condition, we minimized the extent and intensity of threat that would require management or the ecological risk derived from their downstream propagation into freshwater focal areas if not addressed adequately. Moreover, the consideration of some ecological requirements translated into differences not only in the spatial distribution of freshwater focal areas, but also their extent, which helped to further improve the efficiency of conservation recommendations. By accounting for species' mobility, we could address not only the ecological needs of different species, but also reduce the extent of freshwater focal areas. Given the strict protection required in focal areas, it is desirable that their spatial extent is minimized to reduce conservation management costs and potential sociopolitical conflicts.

To aid operationalizing zonal planning in freshwater environments, we provide guidance on how to use some of the key files in Marxan with Zones to adapt the results to the needs of freshwater systems. In particular, we have calibrated the zone boundary file (Table 3; see Watts, Steinback & Klein 2008 for further detail) to arrange spatially the zones and the zone targets file to address species-specific spatial needs (e.g. species of high vs. low mobility). Our recommendations should be taken as guidance on the use of these files and further calibration might be required to adapt the different weights applied here to the special configuration and needs in a given catchment.

CONCLUSIONS

The use of systematic multizone plans could help unlock the potential of conservation recommendations for

freshwater ecosystems (Abell, Allan & Lehner 2007) by making them more informative and explicit in terms of spatial allocation of different management regimes. These new advances should help encourage the implementation of Abell, Allan & Lehner's conservation scheme in a systematic way and enhance the cost-effectiveness of conservation outcomes. The use of cost data or different conservation targets can lead to significant differences in the allocation and extent of zones as we reported here when we used data on threats or included the ecological requirements of species. This has also been reported in previous zonation exercises in marine environments, for example, (see Klein *et al.* 2009) and calls for special caution when defining conservation goals and the role each management zone would play. As we demonstrated here, this could also help optimize the extent of each management zone and enhance the efficiency of conservation recommendations. To help produce ecologically coherent and efficient zoning plans, special attention should be given to developing better informed targets reflecting real conservation needs and estimates of conservation cost.

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Data accessibility

Species descriptions: uploaded as online supporting information (Appendix S1).

Spatial distribution of freshwater fish species in the Daly River catchment and grazing pressure: Dryad Digital Repository: doi.org/10.5061/dryad.f54v6 (Hermoso *et al.* 2015).

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Supporting Information

Additional Supporting Information may be found in the online version of this article.

Appendix S1. List of freshwater fish species included in this study, their distribution area in the Daly River catchment and mobility capacity.